4 Find the solution of the differential equation

$$\sin\theta\frac{\mathrm{d}y}{\mathrm{d}\theta}+y=\tan\tfrac{1}{2}\theta,$$

where $0<\theta<\pi$, given that $y=1$ when $\theta=\frac{1}{2}\pi$. Give your answer in the form $y=\mathrm{f}(\theta)$. [9]

[You may use without proof the result that $\int \mathrm{cosec}\,\theta\,\mathrm{d}\theta=\ln\tan\frac{1}{2}\theta$.]